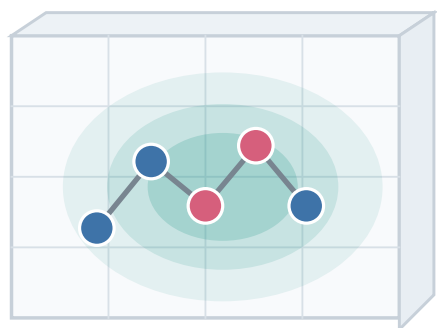


Spatial two-tower pre-training

Electron-density partitioning, independent masked autoencoding, and a frozen-token affinity check

(a) Density partitioning



Aligned complex + 2Fo–Fc ED

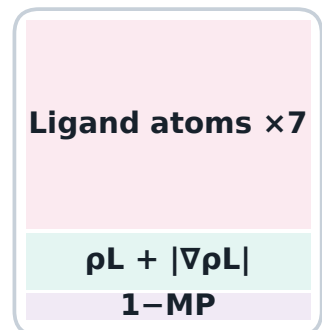
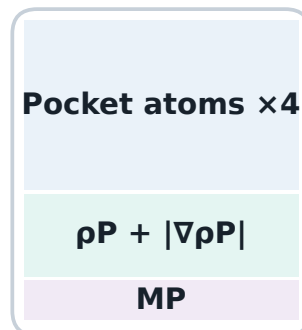
$$M_P = \mathbb{1}[\sum_C V_{P,C} > \tau]$$

$$\rho_P = M_P \odot \rho_{exp}$$

$$\rho_L = (1 - M_P) \odot \rho_{exp}$$

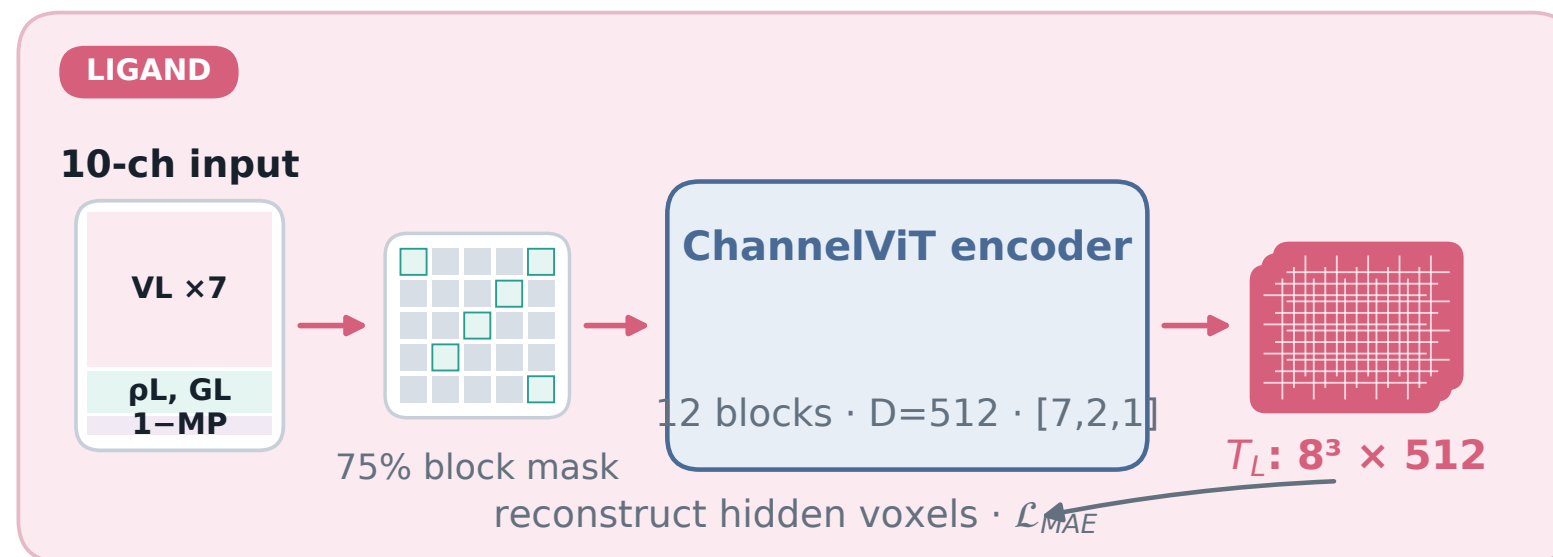
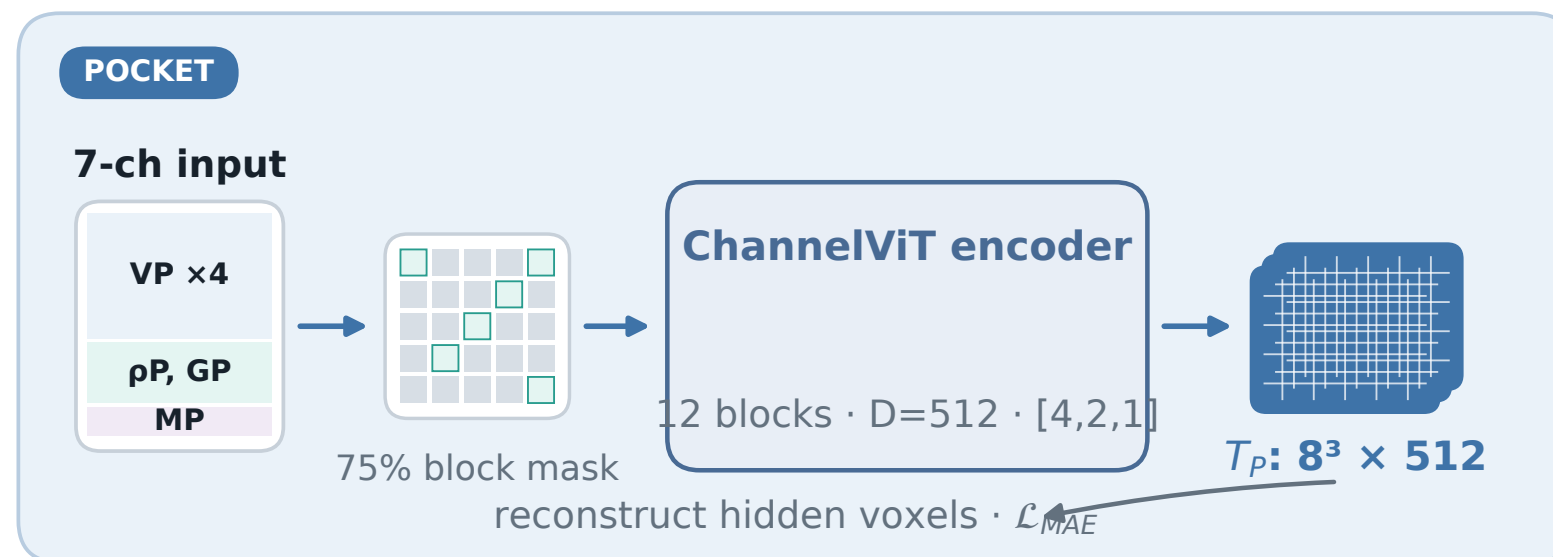
Pocket • 7 ch

Ligand • 10 ch



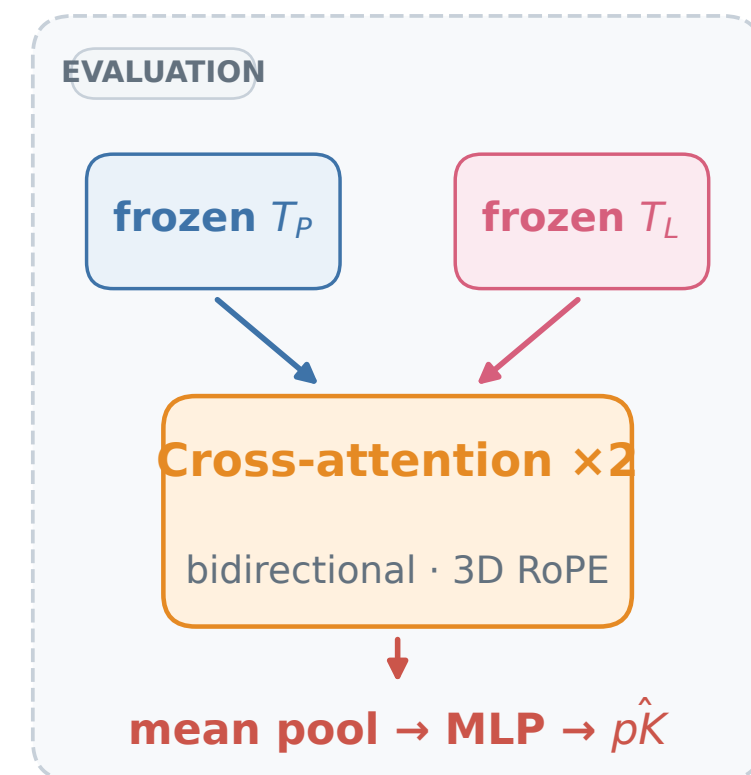
Shared center, orientation, crop, and resolution

(b) Independent MAE towers

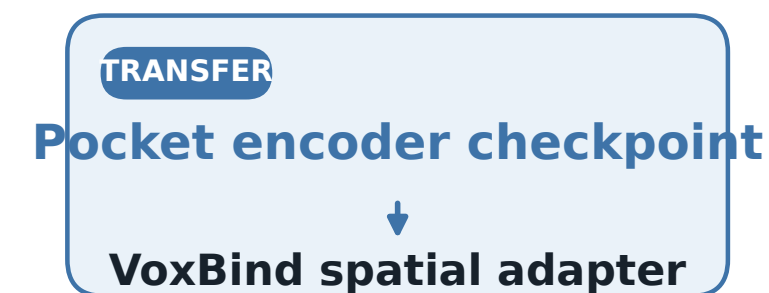


Two separate 50-epoch runs · 112k PLINDER-v2 complexes

(c) Affinity & transfer



The towers remain unchanged



Only the pre-fusion pocket path is reused

■ pocket ■ ligand ■ electron density ■ region mask ■ trainable head ■ frozen module